MD Saib Hossain

AI/ML Engineer | Researcher | LLM & Agentic AI

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Profile

AI/ML Engineer experienced in building and deploying **LLM-powered applications, agentic workflows, machine-learning systems, and full-stack AI products**. Hands-on across the lifecycle from system design and model/API integration to deployment, debugging, evaluation, and maintenance. Strong in **Python, SQL, LLM APIs, RAG, LangGraph, FastAPI, Docker, and cloud deployment**, with research experience across AI healthcare, medical imaging, agentic AI, AI architect, and AI safety.

Research & Professional Experience

1. Research Assistant | Institute of Advanced Research (IAR) (UIU) | Nov 2025 - Present

Lead multidisciplinary, funded research projects across AI in Healthcare, Medical AI, Agentic AI, & AI Security. Providing research direction, technical oversight, project coordination, and researcher mentorship. Mentor and train junior researchers in medical image analysis, machine learning, Deep learning, AI model development, experimental methodology, and scientific research practices.

2. Full-Stack AI Engineer (Contract, Remote), SCMPCR | Sep 2026 - Present

Built and maintained a digital examination and automated certification platform, covering online assessment, evaluation, and certificate-generation workflows. Managed the full development lifecycle, including architecture, implementation, testing, debugging, AWS deployment, and ongoing maintenance with 40+ active users. *Link*

3. Machine Learning Engineer Intern | Fluvo Soft (Remote) | Nov 2025 - Mar 2026

Designed a synthetic face generation pipeline utilizing StyleGAN2 and Stable Diffusion XL (SDXL) to address data scarcity. Developed and deployed an Agentic RAG system to automate workflows for customer support and HR teams. Worked across model integration, retrieval, and agent orchestration.

4. Volunteer Research Assistant | Ruach Tech (London, UK- Remote) | Jan - Mar 2026

Conducted research on AI safety and alignment, analyzed and summarized advanced AI research papers. Supported the development of safe AI system frameworks for ongoing R&D projects.

Working Projects & Services

1. **AI English Conversation & IELTS Practice Agent:** Building an interactive AI English conversation agent for IELTS-style speaking practice. Designed conversational workflows for question generation, contextual interaction, follow-up questions, and response evaluation. Integrated OpenAI APIs with voice interaction and session-level context management.
2. **AI Lung Cancer Screening & Clinical Chatbot:** UIU funded to deploy a web platform integrating AI-driven lung cancer screening with a specialized clinical triage chatbot for preliminary diagnostic guidance. To help the local community and doctors. *link*
3. **Cervical Cancer Screening Prototype:** For non-government organizations, I deployed a cloud-based healthcare screening tool on Render for automated cervical cancer type detection to support rapid medical triage and regional health initiatives. *link*
4. **Educational AI Assistant:** Developed and hosted a low-latency academic chatbot for providing localized, real-time learning support specifically for Bangladeshi students. *link*

AI Learning & Implementation

1. **Agentic AI Technical Interviewer**: Systematically made a non-linear, autonomous AI interviewing agent that dynamically parses PDF resumes to generate tailored technical assessments. Orchestrated complex, stateful conversation flows including real-time answer evaluation, clarifying hints, and adaptive follow-ups. link
2. **ScholarAgent - AI Multi-Agent Scholarship Advisor:** Developed a multi-agent scholarship intelligence system using CrewAI and Ollama with (34–42s) per-agent inference latency and autonomous academic planning workflows. A CrewAI-based agentic AI pipeline for scholarship retrieval, roadmap generation, and SOP automation with real-time web search and (146s) end-to-end execution latency. link
3. **Self-Reflective Retrieval-Augmented Generation (Self-RAG):** Implemented a Self-RAG-inspired pipeline that adaptively retrieves external information and evaluates retrieved content before response generation. link
4. **MedAssist-LLM-iPhone:** Fine-tuned a lightweight *Qwen2.5-0.5B-Instruct LLM* on a medical Question-Answering dataset and optimized it for fully offline inference on an iPhone X. On-Device AI link

Core Technical Skills

Languages: Python, C++, SQL.
ML/ DL: NumPy, Pandas, Scikit-Learn, PyTorch, Vision Transformers, Deep Learning.
LLM/GenAI: OpenAI API, Gemini API, AI Guardrails, Sandbox Testing, AI Governance.
Agentic AI: LangChain, LangGraph, CrewAI, RAG, Multi-Agent Systems, Tool Calling, Stateful Workflows.
Backend & data: FastAPI, REST APIs, PostgreSQL, Supabase
Deployment & LLMOps/Eval: AWS, Docker, Render, Vercel, Cloudflare, LangSmith, DeepEval.
Engineering: System Design, API Integration, Testing, Debugging, Production Maintenance

Education

United International University (UIU) (2022 - 2026)
B.Sc. in Computer Science & Engineering
CGPA: 3.38 out of 4.00

First Author Publications

IEEE ICCIT 2025 (3 Papers):

1. Fracture Classification (ResNet, YOLO, DeepLabV3) — AUC +3.8%, mAP +4.2% *Link*
2. Bangla Sign Language Learning System - 98% accuracy. link
3. Satellite-based Environmental Analysis. link

Additional: 2 Q1-journal manuscripts submitted; ongoing research in Agentic AI security & Cancer.

Achievements & Awards

1st Runner-up – Kaggle Competition (Deep Learning in Computational and Spatial Biology)
Best Abstract Award, same competition (Spatial Biology)

Leadership & Communication Skills

Volunteer at Alo-Bhubon Trust in healthcare & community development initiatives. (2025-Present) Link
UIU Mars Rover Team (UMRT) | Optimized autonomous navigation algorithms (2023 - 2024) Link
General Secretary | National Children's Task Force (NCTF) Satkhira (2014 – 2017) Link

Certifications

Data Science Fundamentals: Python & SQL. link
PyTorch: Techniques and Ecosystem Tools: link
IBM AI Engineering: link
Introduction to Cybersecurity: link
Red Hat System Administration: link
Introduction to LangChain - Python: Link
Project Deep Agents: link
AWS AI Practitioner Challenge: link
Model Context Protocol (MCP): link
Financial Markets: Link
SQL (Basic) Certificate**:** link